Course notes

Circle and functions

On the trigonometric circle, the point identified by an angle x in radians a for coordinates $(\cos x, \sin x)$. For all real x,

$$\cos^2 x + \sin^2 x = 1, \quad -1 \leq \cos x, \sin x \leq 1.$$

Cosine is paired: $\cos(-x) = \cos x$. Sine is odd: $\sin(-x) = -\sin x$. The two functions are: 2π-periodic.

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

Derivatives and variations

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

$$(\sin(ax+b))' = a\cos(ax+b), \quad (\cos(ax+b))' = -a\sin(ax+b).$$

On $[-\pi,\pi]$, cosine grows on $[-\pi,0]$ of -1 to 1, then decreases on $[0,\pi]$ from 1 to -1. Sine decreases on $[-\pi,-\frac{\pi}{2}]$, grows on $[-\frac{\pi}{2},\frac{\pi}{2}]$, then decreases on $[\frac{\pi}{2},\pi]$.

Justification. Read the sign of sine and cosine on the circle and then apply the derivative link·variations. The periodicity prolongs the curves; it does not exempt from studying a length interval sufficient.

Equations and inequities

If $|a|>1$, $\cos x = a$ and $\sin x = a$ no real solution. For a known angle θ,

$$\cos x = \cos\theta \iff x = \theta + 2k\pi \text{ or } x = -\theta + 2k\pi,$$

$$\sin x = \sin\theta \iff x = \theta + 2k\pi \text{ or } x = \pi - \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

The solutions are then restricted to the required range and duplicates are eliminated.

Examples on $[-\pi,\pi]$. $\cos x = \frac{1}{2}$ gives $x = \pm\frac{\pi}{3}$; $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. $\cos x \le \frac{1}{2}$ gives $[-\pi,-\frac{\pi}{3}]\cup[\frac{\pi}{3},\pi]$; $\sin x \ge \frac{1}{2}$ gives $[\frac{\pi}{6},\frac{5\pi}{6}]$.

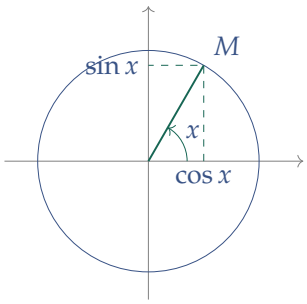
A trigonometric compound

To solve $\cos(2x) = \frac{1}{2}$ on $[-\pi,\pi]$, pose $t = 2x \in [-2\pi,2\pi]$, solve for t, then divide by 2. $x \in \{-\frac{5\pi}{6},-\frac{\pi}{6},\frac{\pi}{6},\frac{5\pi}{6}\}$.

Optimization. For $f(x) = \sin x + \frac{1}{2}\sin(2x)$ on $[0,\pi]$, $f' = \cos x + \cos(2x) = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1)$. The function grows up to $\frac{\pi}{3}$ then decreases; maximum $\frac{3\sqrt{3}}{4}$.

Duplication identities used in exercises. For all real x, $\sin(2x) = 2\sin x\cos x$ and $\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1$. These identities are recalled here so that their employment in problems are not implicit.

Curve symmetries. The cosine parity corresponds to a symmetry with respect to the axis of The periodicity extends to a central symmetry of the original centre. each vector translation curve $(2\pi,0)$.



The coordinates of the point of the circle are $(\cos x, \sin x)$.

From circle to study of function

One angle oriented, two coordinates

First use: S75.

Definition · Sine and cosine

On the unit circle oriented in the direct direction, the point of angle x has for coordinates $(\cos x; \sin x)$. Add 2 $k\pi$ with $k \in \mathbb{Z}$ gives the same point. A negative measure corresponds in the opposite sense; it is not a negative length.

Example Solved · Read an equation and an inequation

On $[-\pi; \pi]$, $\cos x = 1/2$ corresponds to the two points of abscisse 1/2: $x = \pm\pi/3$. Equation $\cos x \leq 1/2$ selects the arc to the left of the vertical right of the abscisse 1/2: $[-\pi; -\pi/3] \cup [\pi/3; \pi]$. The boundaries belong to the solution because inequality is wide.

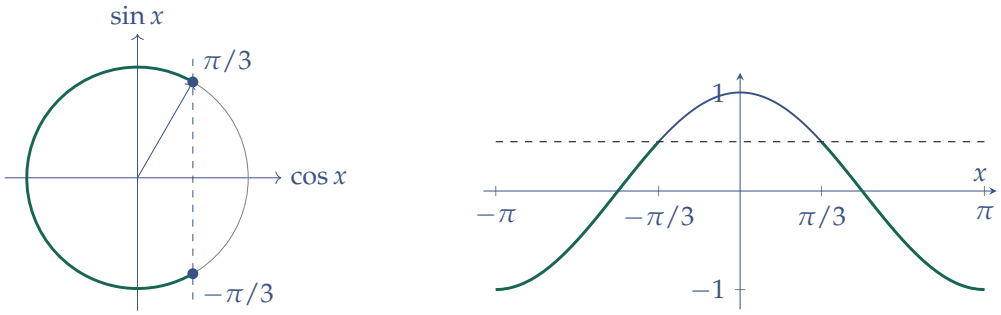


Figure 5.1. Same condition $\cos x \leq 1/2$: arc to the left of the vertical on the circle, curve below level 1 /2 à Right.

A1 · Immediate application Solve $\sin x = 1/2$ on $[0; 2\pi[$then $\sin x \geq 1/2$ on the same Indicate solutions on a circle and curve.

5.2 Read amplitude and period

First use: S76.

Example resolved · A staggered oscillation

For $h(t) = 3 + 2\sin(4t)$, the values are between 1 and 5 : the average level is 3, The smallest positive period is 2 $\pi/4 = \pi/2$. Number 2 π remains one period, but is not the smallest.

A2 · Partial guidance For $g(t) = 1 - 3\cos(2t)$, give minimum, maximum, amplitude and more In order to determine the values of t in $[0; \pi]$ where the extrema are affected.

A3 · Refuter "The smallest positive period of $\sin^2 x$ is 2 π, such as that of $\sin x$. » Show that a translation of π leaves the function unchanged; explain why $\pi/2$ is not a period.

5.3 The sign comes from the circle

First use: S78.

Property · Derivatives and antiderivatives

The derivatives are: $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$. For a compound, multiply by the
A primal of $\cos(ax+b)$ for $a \neq 0$ is $\sin(ax+b)/a$; a antiderivative
Of sin $(ax+b)$ is $-\cos(ax+b)/a$.

Example Solved · Variation over an interval

For $f(x) = x + \cos x$ on $[0;\pi]$, $f' = 1 - \sin x \geq 0$, with equality only in $\pi/2$. The function is strictly increasing: it grows strictly on either side of this isolated point and remains continues. $f'' = -\cos x$ change of sign in $\pi/2$, so the curve has inflection in $(\pi/2;\pi/2)$. A horizontal tangent is not necessarily an extreme.

A4 · Choice of strategy Study $f(x) = x - \sin x$ on $[0;\pi]$: derivative, variations, convexity. $\sin x \leq x$ on this interval. $f'(0) =$ Is it causing a change in variation?

5.4 An identity serving an area

First use: S79.

Example Solved · Area of a rectangle

One point of the first quadrant of the unit circle has for coordinates $(\cos\theta;\sin\theta)$, $0 \leq \theta \leq \pi/2$. The rectangle symmetrical with respect to the axes has as its area $A(\theta) = 4\sin\theta\cos\theta$. Like $(\sin\theta - \cos\theta)^2 \geq 0$ and $\sin^2\theta + \cos^2\theta = 1$, we have $2\sin\theta\cos\theta \leq 1$, therefore $A \leq 2$. Equality requires $\theta = \pi/4$: the optimal rectangle is a square.

A5 · Summary Determine by bypass the maximum of 4 $\sin\theta\cos\theta$ on $[0;\pi/2]$. Find the previous conclusion, then calculate $\int_0^{\pi/2} \sin\theta\cos\theta\,\mathrm{d}\theta$.

A6 · Autonomous problem Demonstrate that $\cos x = x$ has a single solution on $[0;\pi/2]$. Framework between 0.7 and 0.8 using the radian-mode calculator.
precision.

A first step: Study the difference between the two members and separate existence, uniqueness and approximation.

Wait a minute at the courthouse. Explain how the same set of solutions appears as arcs on the circle and intervals on the axis of the apses of the curve.

Review postponed · S84 and then S90 Mixing boundaries, derivation, equations, areas and trigonometry in same study; repeat a resolution over another interval by controlling the bounds.

Research activity

Height of one point on a wheel · 35 min.

A wheel rotates at constant speed in a vertical plane. Its radius is 2 m and its centre is found 3 m from the ground. A point of the wheel is at its lowest position at the moment $t = 0$. Wheel makes a turn in 8 s. Its height is modelled by $h(t) = 3 - 2\cos\left(\frac{\pi t}{4}\right)$ where: t is expressed in seconds and $h(t)$ In meters.

1. Verify that the model gives the initial height announced, a period of 8 s and heights minimum and maximum expected.
2. Calculate $h'(t)$, then study the variations in h on $[0;8]$.
3. Determine every moment $t \in [0;8]$ for which the height is 4 m.
4. Determine the total time during this round during which the point is at least 4 m of the ground.